

Applications of ICT (ICT 101)

Lecture 2: Working with Real Data in Excel

Mr. Abdul Basit
Lab Instructor

School of Data Science and Information Technology
NIT

Fall 2026



Excel Lesson 1: Navigating, Formatting, Sorting & Filtering

Today's Agenda

- ① **Start Here**
- ② **Getting Around**
- ③ **Making It Readable**
- ④ **Conditional Formatting**
- ⑤ **Sort & Filter**
- ⑥ **Today's Lab**
- ⑦ **Wrap-Up**

Meet Today's Data: 1,000 Orders

A1	Order ID																			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	Order ID	Order Date	Customer	City	Province	Category	Product	Quantity	Unit Price	Discount	Payment M	Status	Delivery	De	Rating					
2	ORD-10001	7/1/2025	Hassan Ans	Multan	Punjab	Beauty	Face Wash	6	890	0.15	Cash on De	Delivered	1	5						
3	ORD-10002	7/2/2025	Iqra Quresh	Rawalpindi	Punjab	Home & Kit	Blender	1	10440	0	Card	Delivered	2	3						
4	ORD-10003	7/2/2025	Hamza Mali	Peshawar	Khyber Pak	Electronics	Laptop	5	182200	0	Cash on De	Delivered	7	5						
5	ORD-10004	7/2/2025	Omar Javeed	Islamabad	Islamabad	Clothing	Sneakers	6	7820	0	Cash on De	Delivered	8	3						
6	ORD-10005	7/2/2025	Hamza Cha	Islamabad	Islamabad	Home & Kit	Blender	5	10930	0	Bank Transf	Delivered	2	3						
7	ORD-10006	7/3/2025	Junaid Kha	Quetta	Balochistan	Grocery	Dry Fruit Bl	10	4700	0.1	Cash on De	Delivered	6	5						
8	ORD-10007	7/3/2025	Haris Aslam	Quetta	Balochistan	Beauty	Hair Dryer	2	6770	0.1	Card	Delivered	4	5						
9	ORD-10008	7/4/2025	Shoab Zaf	Islamabad	Islamabad	Beauty	Perfume	5	7060	0.05	Cash on De	Delivered	10	2						
10	ORD-10009	7/4/2025	Billal Javed	Islamabad	Islamabad	Home & Kit	Electric Ket	8	5110	0.1	Cash on De	Delivered	4	3						
11	ORD-10011	7/5/2025	Saima Shah	Lahore	Punjab	Books	Exam Guide	2	1620	0	Mobile Wa	Delivered	7	5						
12	ORD-10011	7/5/2025	Talha Aslam	Karachi	Sindh	Home & Kit	Room Heat	1	15260	0	Card	Delivered	9	4						
13	ORD-10011	7/6/2025	Uman Sidi	Karachi	Sindh	Electronics	Power Banl	1	3790	0.15	Cash on De	Delivered	5	2						
14	ORD-10011	7/7/2025	Billal Malik	Karachi	Sindh	Clothing	Winter Jac	7	9100	0.05	Mobile Wa	Delivered	3	3						
15	ORD-10014	7/7/2025	Zoya Baig	Rawalpindi	Punjab	Home & Kit	Blender	3	10470	0.2	Card	Delivered	7	3						
16	ORD-10011	7/8/2025	Sara Mughi	Karachi	Sindh	Home & Kit	Electric Ket	4	5640	0.05	Cash on De	Pending								
17	ORD-10011	7/8/2025	Yasir Saleem	Lahore	Punjab	Clothing	Lawn Suit	1	4560	0.1	Cash on De	Delivered	4	3						
18	ORD-10011	7/8/2025	Junaid Asla	Peshawar	Khyber Pak	Clothing	Sneakers	5	7960	0	Cash on De	Pending								
19	ORD-10011	7/9/2025	Saad Butt	Sialkot	Punjab	Clothing	Shawl	2	2530	0.05	Card	Delivered	5	4						
20	ORD-10011	7/9/2025	Alishba Ans	Karachi	Sindh	Home & Kit	Blender	5	11060	0	Cash on De	Delivered	4	5						
21	ORD-10024	7/9/2025	Zain Siddiq	Lahore	Punjab	Home & Kit	Bedsheet Si	1	4680	0.15	Mobile Wa	Delivered	1	5						
22	ORD-10021	7/9/2025	Mehwish Al	Peshawar	Khyber Pak	Clothing	Lawn Suit	5	4390	0	Cash on De	Returned	2							
23	ORD-10022	7/9/2025	Usman Razi	Faisalabad	Punjab	Clothing	Winter Jac	2	9700	0.05	Mobile Wa	Delivered	10	5						
24	ORD-10023	7/9/2025	Bushra Hani	Rawalpindi	Punjab	Home & Kit	Bedsheet Si	4	4710	0.1	Cash on De	Delivered	2	3						
25	ORD-10024	7/9/2025	Waqas Iqbal	Quetta	Balochistan	Home & Kit	Blender	2	11990	0	Cash on De	Cancelled								
26	ORD-10022	7/9/2025	Rabia Sheikh	Rawalpindi	Punjab	Home & Kit	Bedsheet Si	2	4740	0	Cash on De	Delivered	1	4						
27	ORD-10021	7/9/2025	Hassan She	Lahore	Punjab	Books	Cookbook	4	2320	0.05	Bank Transf	Delivered	5	5						
28	ORD-10022	7/9/2025	Sameer Mu	Sialkot	Punjab	Grocery	Cooking Oil	2	3440	0.1	Cash on De	Delivered	6	4						

Lab2-dataset.xlsx

- **1,000** online orders, July 2025 to June 2026
- **14** columns: city, product, price, payment, status, rating...
- One **row** is one order, one **column** is one detail

The manager asks

How many Karachi electronics orders?

Which item was the most expensive?

How many orders came back?

Scrolling will not answer these. By 11:35, you will.

Step by Step: Open the File and Save Your Own Copy

- 1 Download **Lab2-dataset.xlsx** from Canvas and open it in Excel
- 2 Click **File** → **Save As**
- 3 In the name box type `RollNo_Lab2` and use your own roll number, for example `23-NIT-0145_Lab2`
- 4 Make sure the type says **Excel Workbook (.xlsx)**, then click **Save**
- 5 At the bottom of the window, click the **+** next to the sheet tab `Orders`
- 6 Double-click the new tab, type `Answers` and press `Enter`

Why we do this first

The original file stays untouched, so you can always download it again.

Your answers go on the **Answers** sheet: question number in column A, your answer in column B.

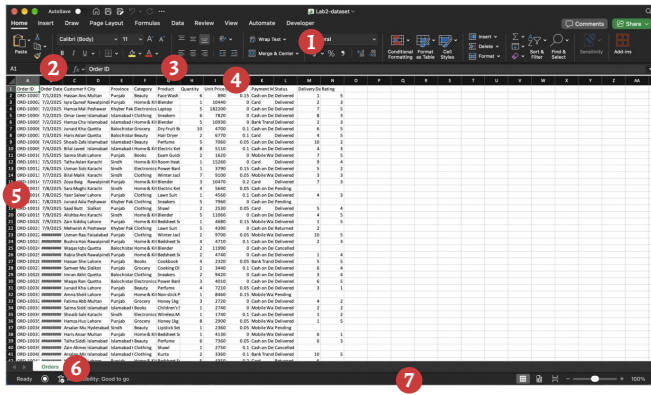
Press `Ctrl + S` after every task.

Orders

Answers

← new tab

The Excel Window: Seven Parts You Will Use Today



- 1 Ribbon:** all commands, sorted into tabs (Home, Data, View)
- 2 Name Box:** shows which cell is selected, and lets you jump to any cell
- 3 Formula Bar:** shows what is really stored in the cell
- 4 Column letters:** A, B, C...
- 5 Row numbers:** 1, 2, 3...
- 6 Sheet tab:** one tab per sheet
- 7 Status bar:** count, sum and average of what you select

A cell's name is its column letter plus its row number. The cell in column G, row 500, is G500.

Step by Step: Jump Around 1,000 Rows with the Keyboard

- 1 Click cell **A1**
- 2 Press **Ctrl + Down Arrow**. You land on the **last order**. Look at the row number on the left
- 3 Press **Ctrl + Right Arrow**. You land on the **last column**. Look at the column letter on top
- 4 Press **Ctrl + Home**. You are back at **A1**
- 5 Press **Ctrl + Left Arrow**, then **Ctrl + Up Arrow** to see how the same trick works in the other directions

The rule behind it

Ctrl + an arrow key runs to the **edge of the filled cells** in that direction. It stops where the data stops.



993	ORD-1099	#####	Eman Khan Lahore	Punjab	Beauty	Lipstick Set	2	2160
994	ORD-1099	#####	Sana Abbas Quetta	Balochistan	Clothing	Sneakers	6	8700
995	ORD-1099	#####	Mehwish R. Faisalabad	Punjab	Grocery	Tea 900g	6	1890
996	ORD-1099	#####	Ayesha Siddi Lahore	Punjab	Grocery	Dry Fruit B	3	4240
997	ORD-1099	#####	Hassan Jave Multan	Punjab	Electronics	Laptop	1	176890
998	ORD-1099	#####	Sara Akhtar Hyderabad	Sindh	Grocery	Cooking Oil	6	3000
999	ORD-1099	#####	Fahad Raza Sialkot	Punjab	Beauty	Perfume	8	6370
1000	ORD-1099	#####	Nida Iqbal Peshawar	Khyber Pak	Grocery	Cooking Oil	1	3070
1001	ORD-1100	#####	Fahad Iqbal Faisalabad	Punjab	Grocery	Honey 1kg	10	2770

After step 2: the Name Box and the row numbers show where you landed. The data ends at row 1001.

Keys	Does
Ctrl + Home	back to A1
Ctrl + End	last used cell
Ctrl + Shift + Down	select down to the end

Step by Step: Use the Name Box to Jump and to Select

A. Go straight to one cell

- 1 Click inside the **Name Box** (top left, above column A)
- 2 Type G500
- 3 Press Enter
- 4 Excel jumps to that cell. Read its content in the Formula Bar

B. Select a block of cells

- 1 Click the Name Box and type L2:L1001
- 2 Press Enter
- 3 The whole Status column, without its header, is selected

Remember

L2:L1001 reads as “from L2 down to L1001”. The colon means **to**. This is the safest way to select a column of data without the header, and we will use it again in Conditional Formatting.

Step by Step: Let the Status Bar Count for You

- 1 Click the **column letter** A. The whole column is selected
- 2 Look at the bottom right of the window. This is the **status bar**
- 3 It shows **Count: 1001**. The header cell is counted too, so 1000 orders plus 1 header
- 4 Now click the column letter of a number column, for example **Quantity**
- 5 The status bar now also shows the **Sum** and the **Average**

What the status bar shows

Count: how many filled cells you selected.

Sum: the total (numbers only).

Average: the mean (numbers only).

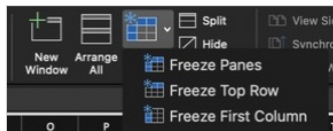
Step by Step: Freeze Panes, Never Lose Your Headers

- 1 Click the **View** tab on the ribbon
- 2 Click **Freeze Panes**
- 3 Choose **Freeze Top Row**
- 4 Scroll down with the mouse wheel or go to row 500
- 5 Row 1 stays at the top, so you can always see what each column means
- 6 To undo it: View → Freeze Panes → **Unfreeze Panes**

Other options: **Freeze First Column** keeps column A in place. **Freeze Panes** keeps everything above and to the left of the selected cell.

G500														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quantity	Unit Price (Rs)	Discount	Payment Method	Status	Delivery Days	Rating
463	ORD-10460	12-Dec-2025	Imran Haider	Lahore	Punjab	Home & Kitchen	Room Heater	2	14,370	0%	Cash on Delivery	Returned		3
462	ORD-10461	12-Dec-2025	Fatima Javed	Islamabad	Islamabad Capital Territory	Beauty	Hair Dryer	1	7,540	15%	Mobile Wallet	Delivered		3
463	ORD-10462	12-Dec-2025	Maryam Qureshi	Hyderabad	Sindh	Clothing	Kurta	1	3,540	5%	Mobile Wallet	Cancelled		4
464	ORD-10463	12-Dec-2025	Alishba Malik	Lahore	Punjab	Clothing	Jeans	1	3,570	10%	Card	Pending		5
465	ORD-10464	14-Dec-2025	Haris Aslam	Lahore	Punjab	Home & Kitchen	Air Fryer	8	25,330	0%	Bank Transfer	Delivered	5	4
466	ORD-10465	14-Dec-2025	Rabia Ahmed	Karachi	Sindh	Home & Kitchen	Blender	2	10,600	5%	Cash on Delivery	Returned	2	5
467	ORD-10466	15-Dec-2025	Iqra Aslam	Lahore	Punjab	Clothing	Kurta	2	2,930	0%	Cash on Delivery	Delivered	4	5
468	ORD-10467	15-Dec-2025	Zain Zafar	Multan	Punjab	Clothing	Jeans	1	3,630	10%	Card	Delivered	1	4
469	ORD-10468	15-Dec-2025	Mahnoor Qureshi	Karachi	Sindh	Grocery	Basmati Rice Skg	6	2,110	10%	Card	Returned	8	5
470	ORD-10469	16-Dec-2025	Amna Raza	Quetta	Balochistan	Books	Children's Story Set	1	2,760	10%	Mobile Wallet	Delivered	2	4
471	ORD-10470	17-Dec-2025	Kiran Ahmed	Karachi	Sindh	Books	Children's Story Set	4	2,590	15%	Card	Delivered	3	4
472	ORD-10471	17-Dec-2025	Laila Raig	Karachi	Sindh	Electronics	Smart Watch	1	13,010	15%	Card	Delivered	4	4
473	ORD-10472	17-Dec-2025	Usman Raza	Peshawar	Khyber Pakhtunkhwa	Home & Kitchen	Bedsheet Set	7	4,180	0%	Bank Transfer	Delivered	3	2
474	ORD-10473	18-Dec-2025	Bushra Raza	Karachi	Sindh	Beauty	Perfume	1	7,580	0%	Mobile Wallet	Delivered	2	4
475	ORD-10474	18-Dec-2025	Bushra Raza	Karachi	Sindh	Electronics	Smart Watch	3	11,840	15%	Mobile Wallet	Delivered	4	4
476	ORD-10475	18-Dec-2025	Iqra Butt	Karachi	Sindh	Clothing	Winter Jacket	1	9,520	0%	Bank Transfer	Returned	7	5
477	ORD-10476	18-Dec-2025	Sana Mirza	Multan	Punjab	Electronics	Smartphone	3	71,250	5%	Mobile Wallet	Delivered	4	4
478	ORD-10477	19-Dec-2025	Imran Raza	Multan	Punjab	Grocery	Tea 900g	12	1,790	15%	Cash on Delivery	Cancelled		5
479	ORD-10478	20-Dec-2025	Shoaib Javed	Peshawar	Khyber Pakhtunkhwa	Books	Exam Guide	1	1,360	15%	Cash on Delivery	Delivered	2	5
480	ORD-10479	21-Dec-2025	Anum Saleem	Peshawar	Khyber Pakhtunkhwa	Electronics	Wireless Mouse	1	1,780	15%	Mobile Wallet	Pending		5
481	ORD-10480	21-Dec-2025	Komal Malik	Islamabad	Islamabad Capital Territory	Beauty	Lipstick Set	7	2,280	0%	Bank Transfer	Cancelled		5
482	ORD-10481	21-Dec-2025	Bushra Baig	Hyderabad	Sindh	Electronics	Smartphone	1	72,310	0%	Cash on Delivery	Delivered	1	4
483	ORD-10482	22-Dec-2025	Wasim Haider	Quetta	Balochistan	Clothing	Jeans	4	3,480	5%	Mobile Wallet	Delivered	5	5
484	ORD-10483	22-Dec-2025	Adnan Raza	Karachi	Sindh	Electronics	Smart Watch	1	12,010	0%	Bank Transfer	Delivered	7	2
485	ORD-10484	22-Dec-2025	Danish Abbasi	Sialkot	Punjab	Clothing	Kurta	7	2,940	5%	Bank Transfer	Delivered	5	3
486	ORD-10485	23-Dec-2025	Iqra Iqbal	Peshawar	Khyber Pakhtunkhwa	Home & Kitchen	Air Fryer	3	22,020	10%	Cash on Delivery	Cancelled		5
487	ORD-10486	24-Dec-2025	Junaid Mirza	Lahore	Punjab	Home & Kitchen	Room Heater	3	15,810	5%	Mobile Wallet	Delivered	3	3

Result of step 5: headers still on screen, row numbers far down.



The menu in steps 2 and 3.

Raw vs Readable: Where We Are Going

As downloaded

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quantity	Unit Price	Discount	Payment Method	Status	Delivery Date	Rating
2	ORD-10001	7/1/2025	Hassan Ansari	Multan	Punjab	Beauty	Face Wash	6	890	0.15	Cash on Delivery	Delivered	1	
3	ORD-10002	7/2/2025	Iqra Qureshi	Rawalpindi	Punjab	Home & Kitchen	Blender	1	10440	0	Card	Delivered	2	
4	ORD-10003	7/2/2025	Hamsa Malik	Peshawar	Khyber Pakhtunkhwa	Electronics	Laptop	5	182200	0	Cash on Delivery	Delivered	7	
5	ORD-10004	7/2/2025	Omar Javed	Islamabad	Islamabad Capital Territory	Clothing	Sneakers	6	7820	0	Cash on Delivery	Delivered	8	
6	ORD-10005	7/2/2025	Hamsa Chaudhry	Islamabad	Islamabad Capital Territory	Home & Kitchen	Blender	5	10930	0	Bank Transfer	Delivered	2	
7	ORD-10006	7/3/2025	Junaid Khan	Quetta	Balochistan	Grocery	Dry Fruit Box	10	4700	0.1	Cash on Delivery	Delivered	6	
8	ORD-10007	7/3/2025	Haris Aslam	Quetta	Balochistan	Beauty	Hair Dryer	2	6770	0.1	Card	Delivered	4	
9	ORD-10008	7/4/2025	Shoaib Zafar	Islamabad	Islamabad Capital Territory	Beauty	Perfume	5	7060	0.05	Cash on Delivery	Delivered	10	
10	ORD-10009	7/4/2025	Bilal Javed	Islamabad	Islamabad Capital Territory	Home & Kitchen	Electric Kettle	8	5110	0.1	Cash on Delivery	Delivered	4	
11	ORD-10010	7/5/2025	Saima Shah	Lahore	Punjab	Books	Exam Guide	2	1620	0	Mobile Wallet	Delivered	7	
12	ORD-10011	7/5/2025	Talha Aslam	Karachi	Sindh	Home & Kitchen	Room Heater	1	15260	0	Card	Delivered	9	
13	ORD-10012	7/6/2025	Usman Siddiqui	Karachi	Sindh	Electronics	Power Bank	1	3790	0.15	Cash on Delivery	Delivered	5	
14	ORD-10013	7/7/2025	Bilal Malik	Karachi	Sindh	Clothing	Winter Jacket	7	9100	0.05	Mobile Wallet	Delivered	3	
15	ORD-10014	7/7/2025	Zoya Baig	Rawalpindi	Punjab	Home & Kitchen	Blender	3	10470	0.2	Card	Delivered	7	

After five formatting steps

	A	B	C	D	E	F	G	H	I	J
1	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quantity	Unit Price (Rs)	Discount (%)
2	ORD-10001	01-Jul-2025	Hassan Ansari	Multan	Punjab	Beauty	Face Wash	6	890	15%
3	ORD-10002	02-Jul-2025	Iqra Qureshi	Rawalpindi	Punjab	Home & Kitchen	Blender	1	10,440	0%
4	ORD-10003	02-Jul-2025	Hamsa Malik	Peshawar	Khyber Pakhtunkhwa	Electronics	Laptop	5	182,200	0%
5	ORD-10004	02-Jul-2025	Omar Javed	Islamabad	Islamabad Capital Territory	Clothing	Sneakers	6	7,820	0%
6	ORD-10005	02-Jul-2025	Hamsa Chaudhry	Islamabad	Islamabad Capital Territory	Home & Kitchen	Blender	5	10,930	0%
7	ORD-10006	03-Jul-2025	Junaid Khan	Quetta	Balochistan	Grocery	Dry Fruit Box	10	4,700	10%
8	ORD-10007	03-Jul-2025	Haris Aslam	Quetta	Balochistan	Beauty	Hair Dryer	2	6,770	10%
9	ORD-10008	04-Jul-2025	Shoaib Zafar	Islamabad	Islamabad Capital Territory	Beauty	Perfume	5	7,060	5%
10	ORD-10009	04-Jul-2025	Bilal Javed	Islamabad	Islamabad Capital Territory	Home & Kitchen	Electric Kettle	8	5,110	10%
11	ORD-10010	05-Jul-2025	Saima Shah	Lahore	Punjab	Books	Exam Guide	2	1,620	0%
12	ORD-10011	05-Jul-2025	Talha Aslam	Karachi	Sindh	Home & Kitchen	Room Heater	1	15,260	0%
13	ORD-10012	06-Jul-2025	Usman Siddiqui	Karachi	Sindh	Electronics	Power Bank	1	3,790	15%
14	ORD-10013	07-Jul-2025	Bilal Malik	Karachi	Sindh	Clothing	Winter Jacket	7	9,100	5%
15	ORD-10014	07-Jul-2025	Zoya Baig	Rawalpindi	Punjab	Home & Kitchen	Blender	3	10,470	20%

Step What

- 1 Make the header row stand out
- 2 Fit the column widths
- 3 Show dates as 01-Jul-2025
- 4 Show prices as 10,440
- 5 Show discounts as 15%

Where

- Row 1, Home tab
- Between column letters
- Column B, Format Cells
- Column I, Comma Style
- Column J, Percent Style

Step 1: Format the Header Row

- 1 Click the row number **1** on the left. The whole header row is selected
- 2 On the **Home** tab click **B** (Bold)
- 3 Click the **Fill Color** bucket and pick a dark color, for example dark blue
- 4 Click the **Font Color A** and pick **white**
- 5 Click **Center** so the headings sit in the middle of the cells
- 6 Click **Wrap Text**. A long heading such as “Unit Price (Rs)” now breaks onto two lines instead of being cut off

Where are these buttons?

All of them are in the **Font** and **Alignment** groups of the **Home** tab.

If you cannot find one, hover your mouse over the icons. Excel shows the name of each button.

Check yourself

Row 1 is dark with white bold text, and every heading is centered and fully readable.

Step 2: AutoFit the Column Widths

- 1 Click the column letter **A**
- 2 Hold `Shift` and click the column letter **N**. Columns A to N are all selected
- 3 Move your mouse to the line between **any two** column letters. The pointer changes to a double arrow
- 4 **Double-click**
- 5 Every column grows or shrinks to fit its longest entry

If you see

That does not mean something is broken. The column is too narrow to show the number. Double-click the border again and the number appears.

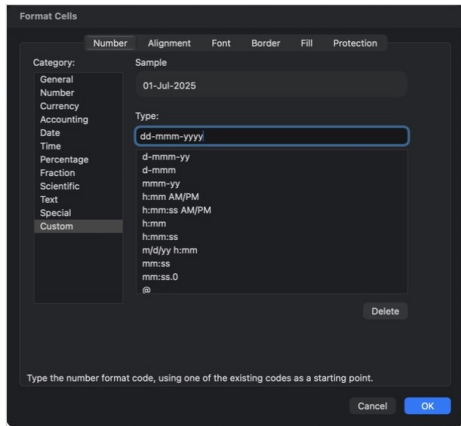
Shortcut

`Ctrl + A` selects the whole sheet, so you can AutoFit everything in one go.

Step 3: Show the Dates in a Cleaner Format

- 1 Click the column letter **B** (Order Date)
- 2 Press **Ctrl + 1**. The **Format Cells** window opens
- 3 On the **Number** tab choose **Custom** in the list on the left
- 4 Click in the **Type** box, delete what is there and type `dd-mmm-yyyy`
- 5 Look at the **Sample** line above. It should read `01-Jul-2025`
- 6 Click **OK**

`dd` = day, `mmm` = month in letters, `yyyy` = four digit year



Steps 4 and 5: Prices with Commas, Discounts as Percentages

Step 4: Unit Price (column I)

- 1 Click the column letter **I**
- 2 On the Home tab, in the **Number** group, click the **comma** button (Comma Style)
- 3 Prices now read 10,440.00
- 4 Click **Decrease Decimal** (the .00 with a left arrow) two times
- 5 Prices now read 10,440

Step 5: Discount (column J)

- 1 Click the column letter **J**
- 2 On the Home tab, in the **Number** group, click the **%** button (Percent Style)
- 3 0.15 now reads 15% and 0 reads 0%

Check yourself

Click any date, price and discount cell. Then look at the Formula Bar. What do you see there? The next slide explains it.

Value vs Display: The Most Important Idea Today

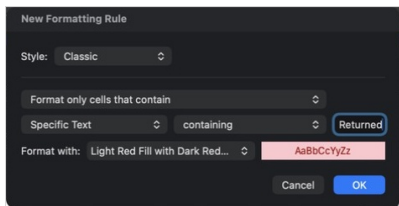
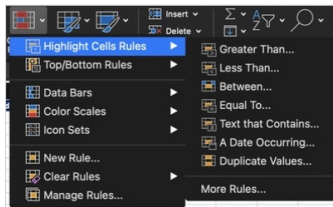
I3											
	A	B	C	D	E	F	G	H	I	J	
1	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quantity	Unit Price (Rs)	Discount	Payment
2	ORD-10001	01-Jul-2025	Hassan Ansari	Multan	Punjab	Beauty	Face Wash	6	890	15%	Cash on Delivery
3	ORD-10002	02-Jul-2025	Iqra Qureshi	Rawalpindi	Punjab	Home & Kitchen	Blender	1	10,440	0%	Card
4	ORD-10003	02-Jul-2025	Hamza Malik	Peshawar	Khyber Pakhtunkhwa	Electronics	Laptop	5	182,200	0%	Cash on Delivery
5	ORD-10004	02-Jul-2025	Omar Javed	Islamabad	Islamabad Capital Territory	Clothing	Sneakers	6	7,820	0%	Cash on Delivery
6	ORD-10005	02-Jul-2025	Hamza Chaudhry	Islamabad	Islamabad Capital Territory	Home & Kitchen	Blender	5	10,930	0%	Bank Transfer

Stored in the cell	Format you applied	Cell shows
10440	Comma Style, no decimals	10,440
0.15	Percent Style	15%
45839	Date, dd-mmm-yyyy	01-Jul-2025

- 1 Click the cell. The **Formula Bar** shows the real value
- 2 Formatting only changes how it **looks**, never what is stored
- 3 Excel keeps dates as numbers (days counted from 1 January 1900). That is why dates can be sorted and filtered

Rule 1: Highlight Every Returned Order in Red

- 1 Click the **Name Box**, type `L2:L1001` and press `Enter`. Only the Status cells are selected, the header is left out
- 2 Home tab → **Conditional Formatting**
- 3 **Highlight Cells Rules** → **Text that Contains...**
- 4 In the box on the left type `Returned`
- 5 In the drop-down on the right choose **Light Red Fill with Dark Red Text**
- 6 Click **OK**



The menu in step 3, and the dialog in steps 4 to 6.

Rules 2 and 3: Expensive Items and Slow Deliveries

Rule 2: Unit Price above 100,000

- 1 Name Box: type I2:I1001, press Enter
- 2 Home → Conditional Formatting → Highlight Cells Rules → **Greater Than...**
- 3 Type 100000
- 4 Choose **Green Fill with Dark Green Text**, click **OK**

Rule 3: Data Bars on Delivery Days

- 1 Name Box: type M2:M1001, press Enter
- 2 Home → Conditional Formatting → **Data Bars**
- 3 Pick any color under **Gradient Fill**
- 4 Longer bar means more days to deliver

Taking a rule away again

Select the range, then Home → Conditional Formatting → **Clear Rules** → **Clear Rules from Selected Cells**.

What You Should See After the Three Rules

Category	Product	Quantity	Unit Price (Rs)	Discount	Payment Method	Status	Delivery Days	Rating
Beauty	Face Wash	6	890	15%	Cash on Delivery	Delivered	1	5
Home & Kitchen	Blender	1	10,440	0%	Card	Delivered	2	3
Electronics	Laptop	5	182,200	0%	Cash on Delivery	Delivered	7	5
Clothing	Sneakers	6	7,820	0%	Cash on Delivery	Delivered	8	3
Home & Kitchen	Blender	5	10,930	0%	Bank Transfer	Delivered	2	3
Grocery	Dry Fruit Box	10	4,700	10%	Cash on Delivery	Delivered	6	5
Beauty	Hair Dryer	2	6,770	10%	Card	Delivered	4	5
Beauty	Perfume	5	7,060	5%	Cash on Delivery	Delivered	10	2
Home & Kitchen	Electric Kettle	8	5,110	10%	Cash on Delivery	Delivered	4	3
Books	Exam Guide	2	1,620	0%	Mobile Wallet	Delivered	7	5
Home & Kitchen	Room Heater	1	15,260	0%	Card	Delivered	9	4
Electronics	Power Bank	1	3,790	15%	Cash on Delivery	Delivered	5	2
Clothing	Winter Jacket	7	9,100	5%	Mobile Wallet	Delivered	3	3
Home & Kitchen	Blender	3	10,470	20%	Card	Delivered	7	3
Home & Kitchen	Electric Kettle	4	5,640	5%	Cash on Delivery	Pending		
Clothing	Lawn Suit	1	4,560	10%	Cash on Delivery	Delivered	4	3
Clothing	Sneakers	5	7,960	0%	Cash on Delivery	Pending		
Clothing	Shawl	2	2,530	5%	Card	Delivered	5	4
Home & Kitchen	Blender	5	11,060	0%	Cash on Delivery	Delivered	4	5
Home & Kitchen	Bedsheet Set	1	4,680	15%	Mobile Wallet	Delivered	1	5
Clothing	Lawn Suit	5	4,390	0%	Cash on Delivery	Returned	2	
Clothing	Winter Jacket	2	9,700	5%	Mobile Wallet	Delivered	10	5

- **Red cells:** Status is Returned
- **Green cells:** Unit Price is above 100,000
- **Bars:** the longer the bar, the more Delivery Days

Nothing was changed

Conditional formatting only paints the cells. Values stay exactly as they were, so you can still sort and filter.

Sorting: The Mistake Almost Everyone Makes

Task: sort these orders by **Unit Price**, smallest to largest

Before

Product	Price
Laptop	165,000
Face Wash	950
Air Fryer	24,000

Only Price column selected

Product	Price
Laptop	950
Face Wash	24,000
Air Fryer	165,000

A laptop for Rs 950. The data is ruined.

Whole table sorted

Product	Price
Face Wash	950
Air Fryer	24,000
Laptop	165,000

Every row moves together.

The safe way

Click **one cell inside the data** and never select a single column. If Excel shows a **Sort Warning**, choose **Expand the selection**.

Step by Step: Sort by One Column

- 1 Click **one cell inside the data**, for example **C5**
- 2 Click the **Data** tab, then **Sort**
- 3 Tick **My data has headers** (top right of the window)
- 4 Under **Column** choose **Unit Price (Rs)** in the **Sort by** drop-down
- 5 Leave **Sort On** as **Cell Values**
- 6 Under **Order** choose **Largest to Smallest**
- 7 Click **OK**
- 8 Row 2 now holds the most expensive item. Read its Order ID and City across the row

Quick sort

Click a cell in the column you want and use the **A→Z** or **Z→A** buttons on the Data tab. Good for one column. For more than one column use the Sort window.

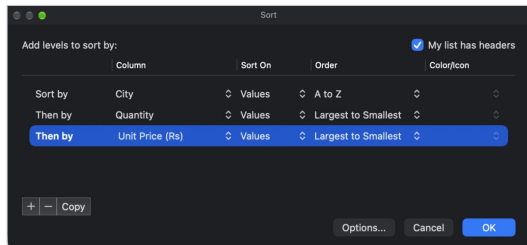
Put things back

Sort by **Order ID**, **Smallest to Largest** (A to Z), and the data is back in its original order. The first row shows ORD-10001 again.

Step by Step: Sort on More Than One Level

- 1 Click one cell inside the data
- 2 Data → **Sort**
- 3 First level: **Sort by City, Sort On Cell Values, Order A to Z**
- 4 Click **Add Level**. Second level: **Then by Quantity, Largest to Smallest**
- 5 Click **Add Level** again. Third level: **Then by Unit Price (Rs), Largest to Smallest**
- 6 Click **OK**

How Excel reads it: first group by City. Inside each city order by Quantity. If two rows still tie, Unit Price decides.



The finished dialog with three levels.

What the Three-Level Sort Looks Like

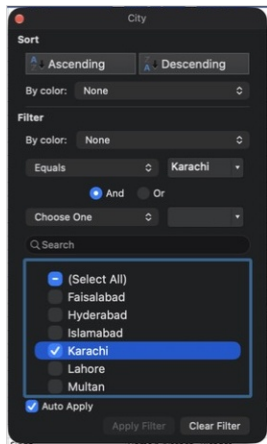
A2 ✕ ✓ f_x ORD-10817														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quantity	Unit Price (Rs)	Discount	Payment Method	Status	Delivery Days	Rating
2	ORD-10817	29-Apr-2026	Faisal Aslam	Faisalabad	Punjab	Grocery	Dry Fruit Box	11	4,640	0%	Cash on Delivery	Pending		
3	ORD-10336	25-Oct-2025	Haris Malik	Faisalabad	Punjab	Grocery	Dry Fruit Box	11	4,160	5%	Cash on Delivery	Delivered		5 1
4	ORD-10518	08-Jan-2026	Junaid Chaudhry	Faisalabad	Punjab	Grocery	Dry Fruit Box	10	4,930	10%	Bank Transfer	Delivered		2 5
5	ORD-10755	06-Apr-2026	Rehan Nawaz	Faisalabad	Punjab	Grocery	Honey 1kg	10	3,090	0%	Mobile Wallet	Delivered		7 2
6	ORD-11000	30-Jun-2026	Fahad Iqbal	Faisalabad	Punjab	Grocery	Honey 1kg	10	2,770	0%	Cash on Delivery	Delivered		3 5
7	ORD-10372	08-Nov-2025	Talha Qureshi	Faisalabad	Punjab	Grocery	Basmati Rice 5kg	9	2,490	0%	Mobile Wallet	Returned		6
8	ORD-10449	06-Dec-2025	Sara Akhtar	Faisalabad	Punjab	Clothing	Sneakers	8	9,200	15%	Bank Transfer	Delivered		10 2
9	ORD-10786	19-Apr-2026	Kamran Javed	Faisalabad	Punjab	Grocery	Dry Fruit Box	8	5,040	10%	Bank Transfer	Delivered		3

- 1 All rows for one city sit together. The cities run in alphabetical order, so **Faisalabad** comes first
- 2 Inside one city the Quantity column goes from big to small
- 3 Rows with the same city and the same quantity are ordered by Unit Price
- 4 Compare the top rows of your sheet with this picture to check your three levels

Before the next part

Sort by **Order ID**, A to Z once more. That puts the sheet back in its original order.

Step by Step: Filter by Text



- 1 Click one cell inside the data
- 2 Data tab → **Filter** (shortcut `Ctrl + Shift + L`). A small arrow appears on every header
- 3 Click the arrow on **City**
- 4 Click (**Select All**) once to untick everything
- 5 Tick **Karachi** only and click **OK**
- 6 Click the arrow on **Category**, untick (Select All), tick **Electronics**, click **OK**
- 7 Only the Karachi electronics orders remain

The picture shows the drop-down you see in step 3. The list of values is what is in that column.

Reading the Answer and Clearing the Filter

ORD-10817														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Order ID	Order Date	Customer Name	City	Province	Category	Product	Quant	Unit Price	Discoun	Payment Meth	Status	Delivery D	Rate
283	ORD-10605	04-Feb-2026	Saad Shah	Karachi	Sindh	Electronics	Smartphone	5	71,240	0%	Card	Delivered	4	1
284	ORD-10654	25-Feb-2026	Hanma Haider	Karachi	Sindh	Electronics	Power Bank	4	4,370	15%	Bank Transfer	Delivered	4	1
313	ORD-10624	11-Feb-2026	Arslan Chaudhry	Karachi	Sindh	Electronics	LED Monitor 24 inch	3	37,980	5%	Cash on Delivery	Delivered	5	4
334	ORD-10810	27-Apr-2026	Mehwish Shah	Karachi	Sindh	Electronics	LED Monitor 24 inch	3	37,710	15%	Cash on Delivery	Delivered	4	4
350	ORD-10764	10-Apr-2026	Aad Butt	Karachi	Sindh	Electronics	Smart Watch	3	13,930	0%	Cash on Delivery	Returned	7	7
370	ORD-10474	18-Dec-2025	Bushra Rana	Karachi	Sindh	Electronics	Smart Watch	3	11,840	15%	Mobile Wallet	Delivered	4	4
382	ORD-10538	13-Jan-2026	Maryam Javed	Karachi	Sindh	Electronics	Bluetooth Speaker	3	6,840	0%	Mobile Wallet	Delivered	3	4
393	ORD-10685	11-Mar-2026	Amna Malik	Karachi	Sindh	Electronics	Power Bank	3	4,580	10%	Cash on Delivery	Delivered	2	3
415	ORD-10143	16-Aug-2025	Hassan Shah	Karachi	Sindh	Electronics	Power Bank	3	4,210	0%	Card	Delivered	8	4
437	ORD-10608	05-Feb-2026	Bilal Nawaz	Karachi	Sindh	Electronics	Power Bank	3	3,920	0%	Card	Delivered	3	4
467	ORD-10931	08-Jun-2026	Ayesha Qureshi	Karachi	Sindh	Electronics	Laptop	2	173,760	5%	Card	Delivered	4	4
488	ORD-10251	19-Sep-2025	Areeba Chaudhry	Karachi	Sindh	Electronics	Laptop	2	154,890	5%	Mobile Wallet	Delivered	7	4
549	ORD-10676	08-Mar-2026	Zoya Khan	Karachi	Sindh	Electronics	Smartphone	2	77,030	10%	Mobile Wallet	Delivered	2	3
560	ORD-10047	18-Jul-2025	Rohan Nawaz	Karachi	Sindh	Electronics	Smartphone	2	69,960	5%	Mobile Wallet	Pending		
551	ORD-10141	15-Aug-2025	Imran Khan	Karachi	Sindh	Electronics	Smartphone	2	69,650	5%	Bank Transfer	Delivered	4	4
552	ORD-10940	11-Jun-2026	Mehwish Saleem	Karachi	Sindh	Electronics	Smartphone	2	66,860	15%	Mobile Wallet	Delivered	2	4
553	ORD-10396	18-Nov-2025	Omar Chaudhry	Karachi	Sindh	Electronics	Smartphone	2	66,000	15%	Mobile Wallet	Delivered	5	2
554	ORD-10918	03-Jun-2026	Junaid Ansari	Karachi	Sindh	Electronics	Smartphone	2	64,930	10%	Mobile Wallet	Delivered	3	2
Ready 56 of 1000 records found Accessibility: Good to go														

Two filters on at once: City and Category. Notice the blue row numbers and the funnel on the filtered headers.

- 1 Look at the bottom left of the window. Excel writes **x of 1000 records found**
- 2 That **x** is the answer to “how many?”
- 3 Rows are **hidden**, not deleted. Blue row numbers tell you so
- 4 To start a new question: Data → **Clear**

No message?

Click the first visible Order ID, press **Ctrl + Shift + Down** and read **Count** in the status bar. Hidden rows are not counted.

Step by Step: Filters for Numbers, Dates and Colors

Numbers

- 1 Arrow on **Quantity**
- 2 **Number Filters** → **Greater Than Or Equal To...**
- 3 Type 10, click **OK**

Dates

- 1 Arrow on **Order Date**
- 2 Untick (**Select All**)
- 3 Click the + next to **2025**, then tick only **December**
- 4 Click **OK**

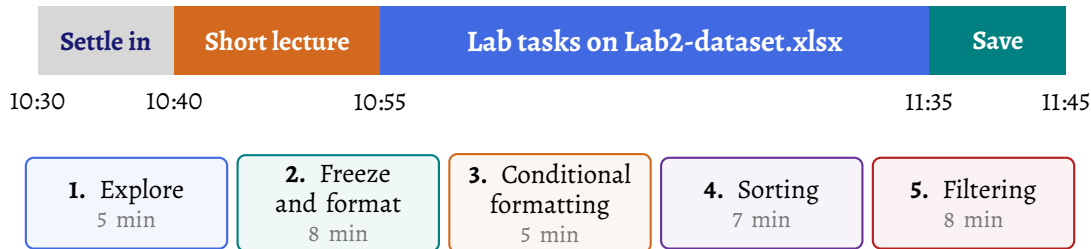
Colors

- 1 Arrow on **Status**
- 2 **Filter by Color** → **Filter by Cell Color**
- 3 Click the light red box that matches your Rule 1

Three habits that save marks

1. Clear the old filter before starting a new question (Data → Clear).
2. Read the answer from the message at the bottom left.
3. Write the answer on the Answers sheet straight away, then go to the next question.

Today's Lab: Lab Manual 2



- **Stuck?** Come back to the step-by-step slides in this lecture. They have every step
- **Save as** RollNo_Lab2.xlsx and write your answers on the **Answers** sheet

How the lab manual works

Each task tells you **what** to produce, not how to click. The how is on the step-by-step slides you just saw. Do the tasks in order, they get harder.

Open Lab2-dataset.xlsx and start Task 1.



ICT 101: Applications of ICT Lab

Try first, then ask. I will be walking around the lab.